

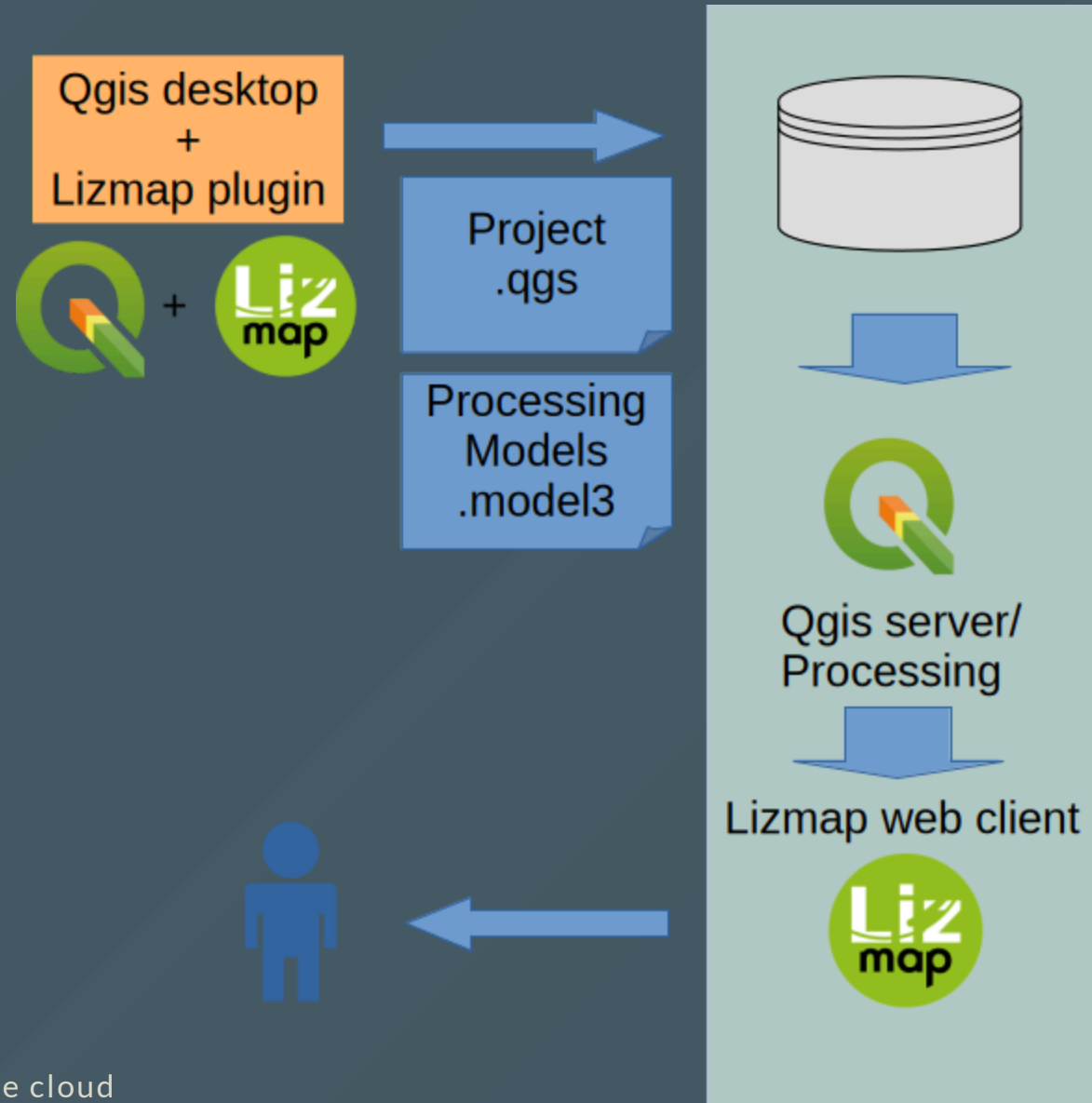
QJazz



QGIS server solution for the cloud

Return of experience from GIS hosting services with Lizmap and QGIS server

GIS hosting services with Lizmap and QGIS server



GIS hosting services with Lizmap and QGIS server

Numbers:

- 580 QGIS server instances
- 1500 projects
- 47 physical servers
- 407 PostgreSQL databases
- 377 lizmap instances



Problems to solve:

- Scalability
- Distributed architecture
- Monitoring
- Zeroconf (ideally !)
- Security

Dealing with issues

- Projects with many layers: up to 200 layers per project
- Loading times of several minutes
- Memory issues
- High latency requests (mainly remote services)
- Stuck server instances

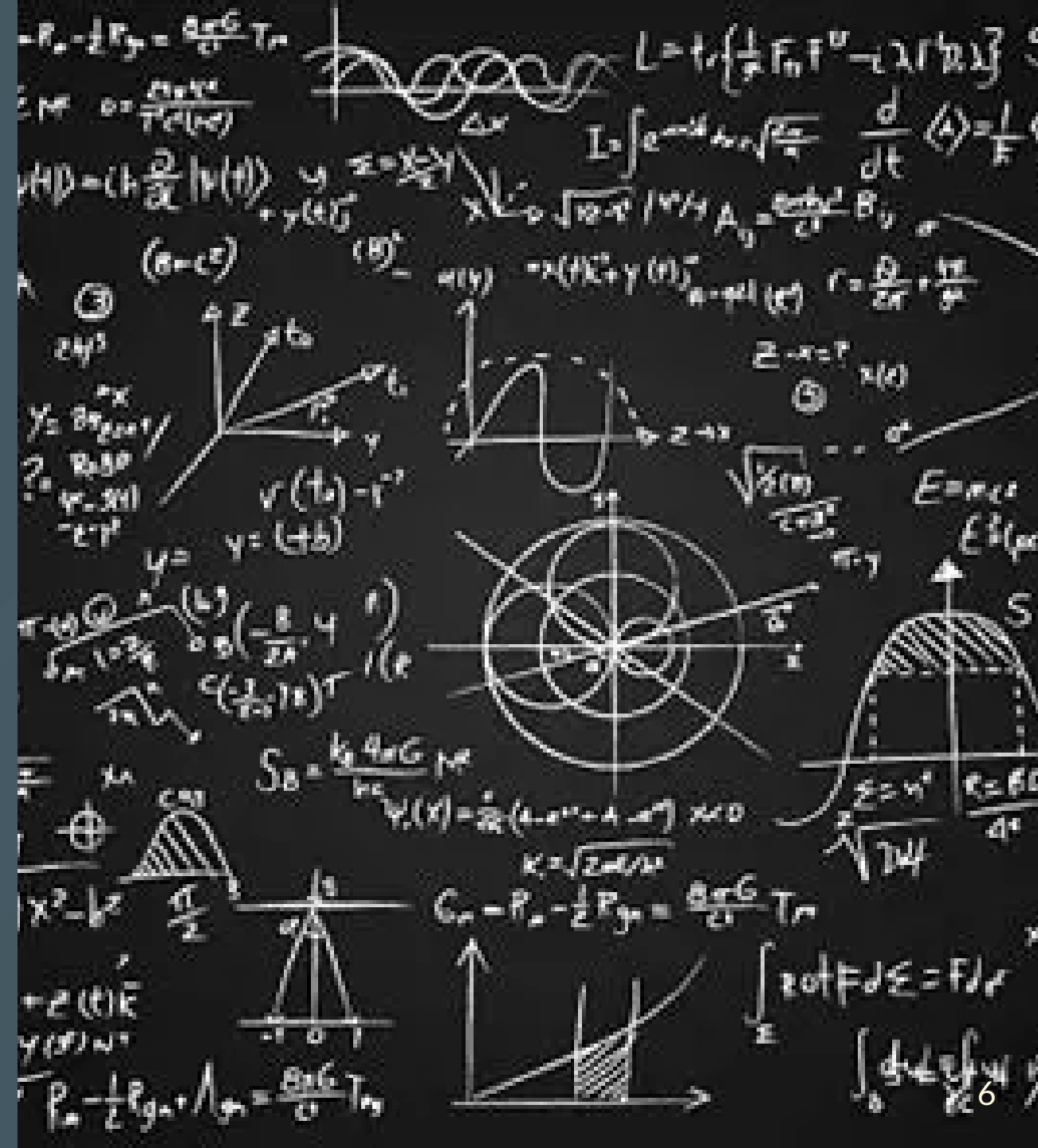
When things go wrong



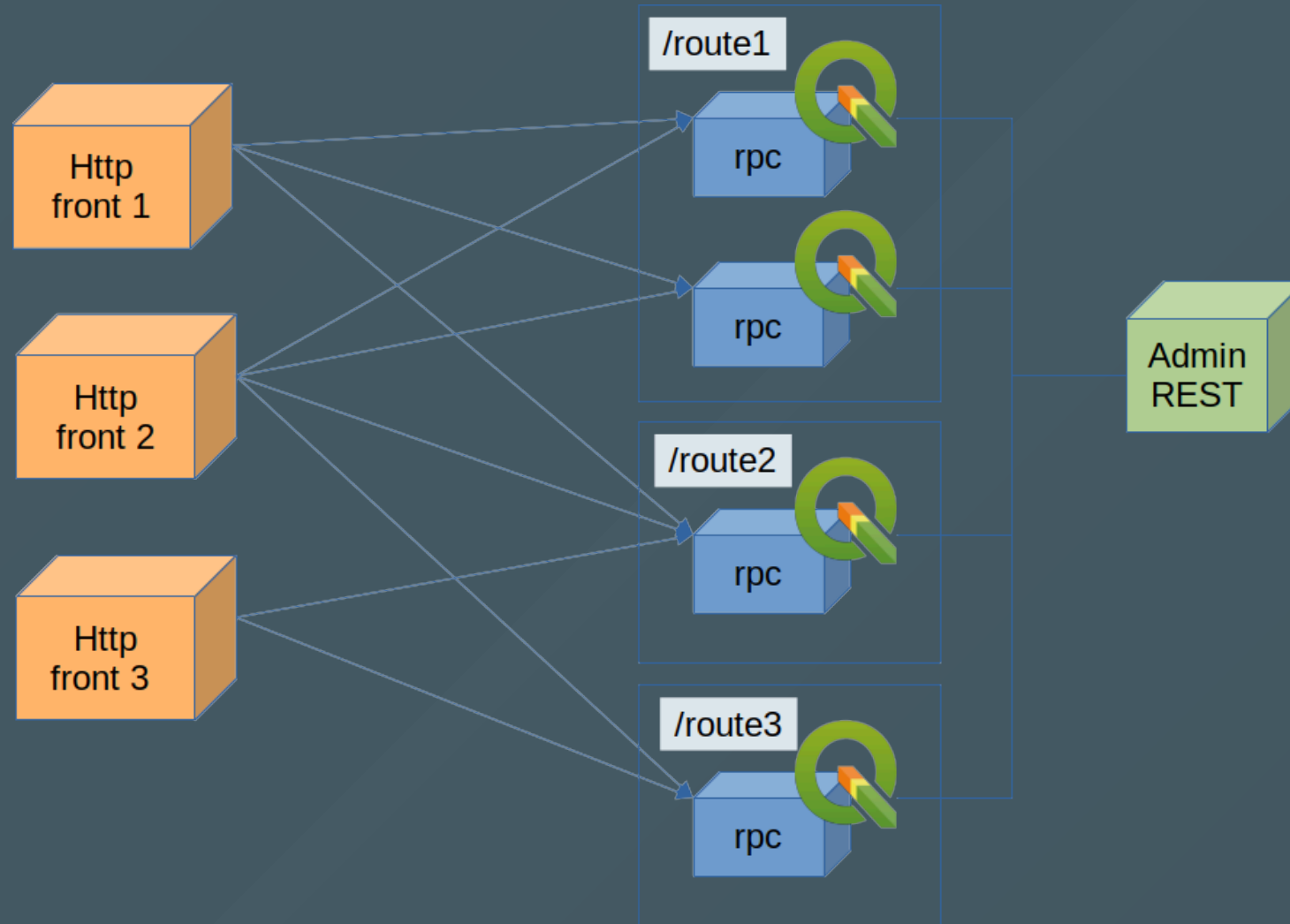
Microservices with gRPC protocol:

<https://github.com/3liz/qjazz>

- Built-in Load balancing
- Built-in healthcheck support
- Dedicated administration service
- Bi-directional streaming support
- Independent from HTTP front-end



QJazz overview



QJazz services

- OGC OWS services (WMS, WFS, ...) - OGIS server native services
- STAC catalogs view of Projects and layers
- OGC API Maps - <https://ogcapi.ogc.org/maps/>
- OGC API Processes - QGIS Processing as a service
- Support for S3 backend (Projects and data)

QGIS Projects management in QJazz

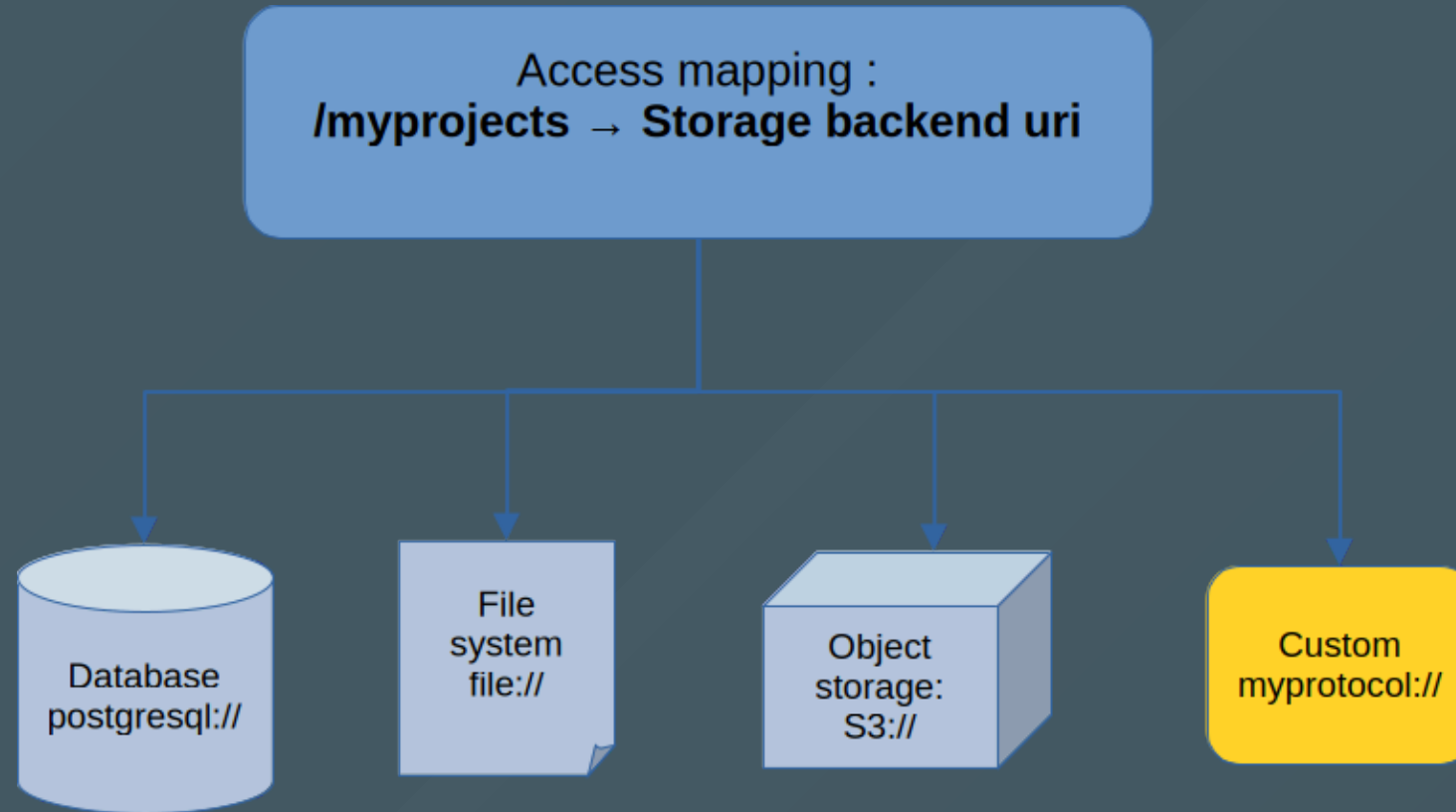
From a 'project as resource' perspective

- Consider Project as an application
- Corollary: QGIS server is an application server
- Control what is published (from a customer perspective)
- Keep some level of flexibility (dynamic caching)

From a deployment perspective

- Projects partitioning (routes)
- Handle access controls to backend Apis
- Easy to scale

Qjazz project's storage



Performances considerations

Horizontal scalability governance

- What is the expected request rate ?
- How request distribute on projects ?
- How many different projects I have to handle ?

Impact on request processing

- How many layers in my project ?
- Data volumetry ?
- Accesses to database backends/external services ?

There is no single strategy to rule all your projects



Conclusion

- Revisiting QGIS server as an application server
- Cloud friendly integration (QGIS servers as micro-services)
- Address the problem of serving multiple use cases with many different kind of QGIS projects.

Thank you



Questions ?